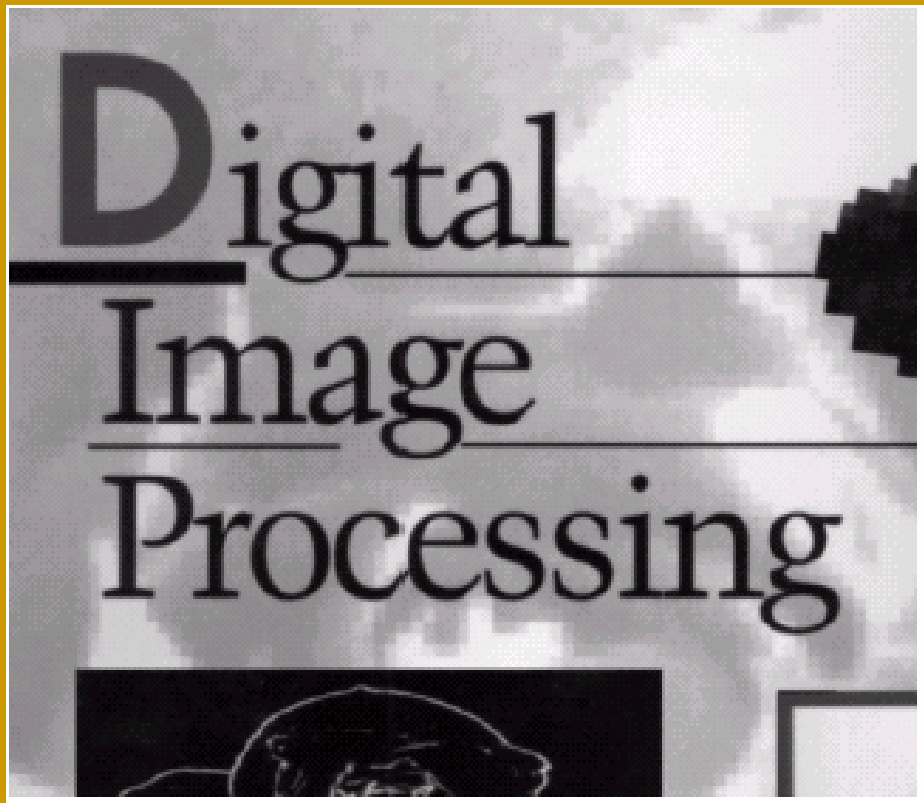
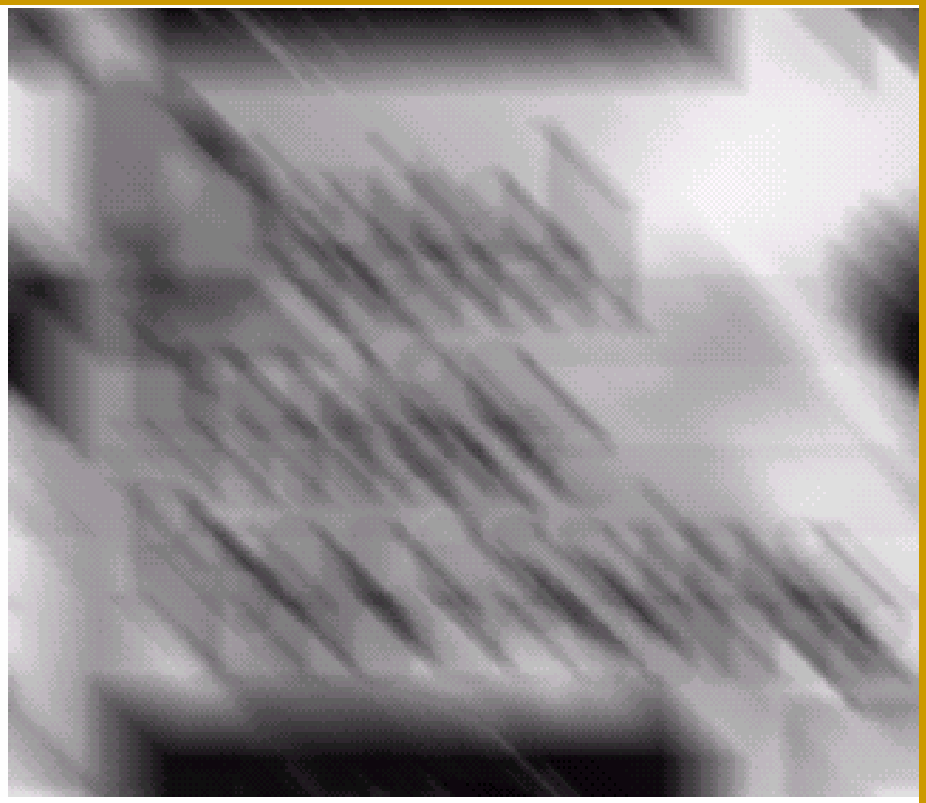




Digital Image Processing
Lecture 8: Image Restoration

Dr.Wafaa Shalash

Review

- In last lecture, we discussed techniques that restore images in spatial domain.
  - Mean filtering
  - Order-statistics filering
  - Adaptive filering
  - Gaussian smoothing
- We'll discuss techniques that work in the frequency domain.

Periodic Noise Reduction

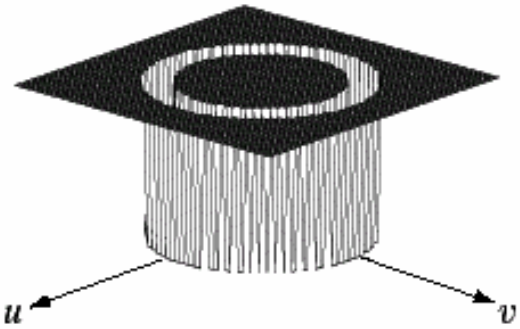
- We have discussed low-pass and high-pass frequency domain filters for image enhancement.
- We'll discuss 2 more filters for periodic noise reduction
  - Bandreject
  - Notch filter



Bandreject Filters

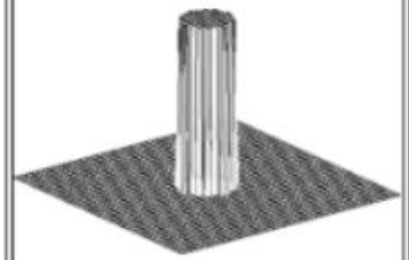
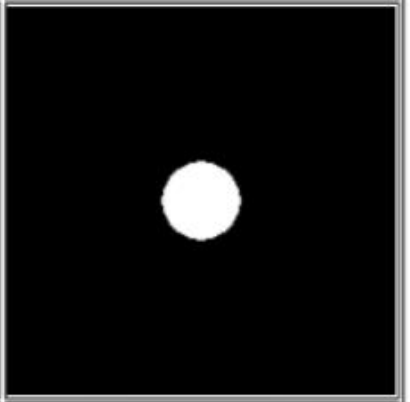
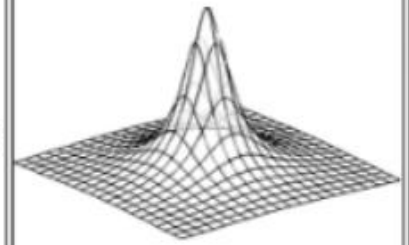
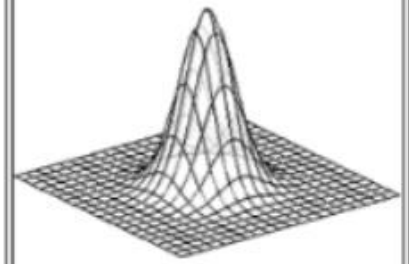
- Removing a band of frequencies about the origin of the Fourier transform.
  - Ideal filter

$$H(u,v)=\begin{cases}1 & \text{if } D(u,v)<D_0-\dfrac{W}{2}\\ 0 & \text{if } D_0-\dfrac{W}{2}\le D(u,v)\le D_0+\dfrac{W}{2}\\ 1 & \text{if } D(u,v)>D_0+\dfrac{W}{2}\end{cases}$$

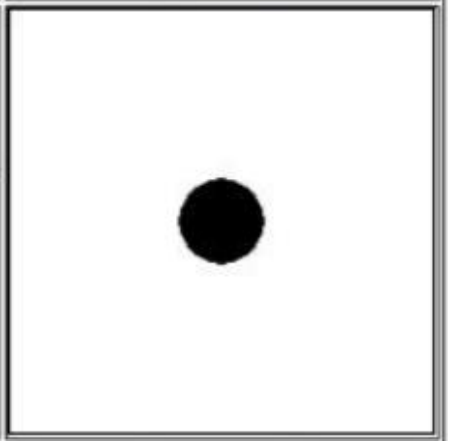
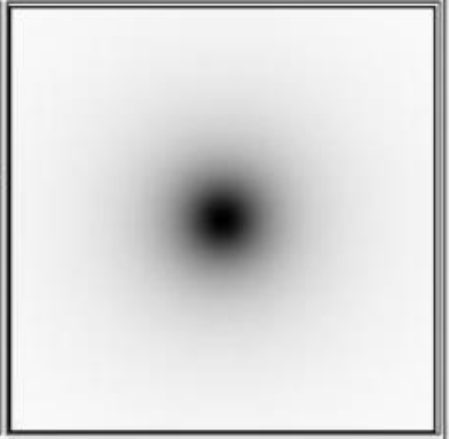
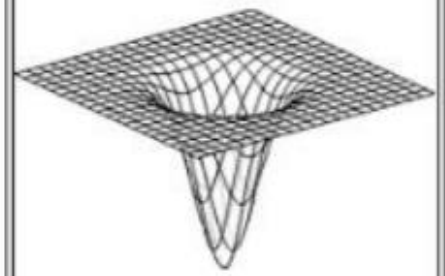
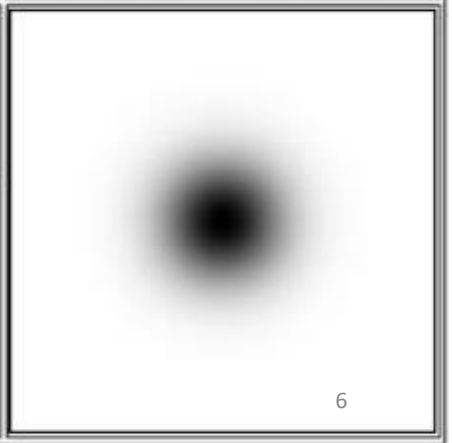


where D(u,v) is the distance from the center, W is the width of the band, and D_0 is the radial center.



Lowpass Filter	Mesh	Image
Ideal: $H(u,v) = \begin{cases} 1 & \text{if } D(u,v) \le D_0 \\ 0 & \text{if } D(u,v) > D_0 \end{cases}$		
Butterworth: $H(u,v) = \dfrac{1}{1+\left[D(u,v)/D_0\right]^{2n}}$		
Gaussian: $H(u,v) = e^{-D^2(u,v)/2D_0^2}$		

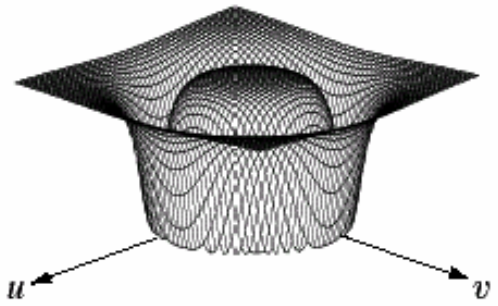


Highpass	Mesh	Image
Ideal		
Butterworth		
Gaussian		

Bandreject Filters (con'd)

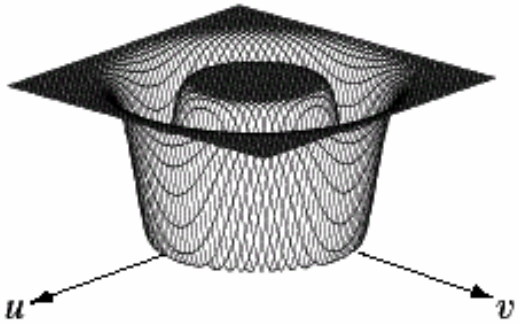
- Butterworth filter of order *n*

$$H(u,v)=\frac{1}{1+\left[\dfrac{D(u,v)W}{D^2(u,v)-D_0^2}\right]^{2n}}$$



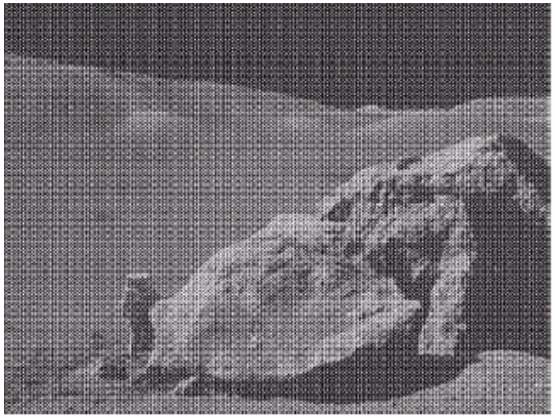
- Gaussian filter

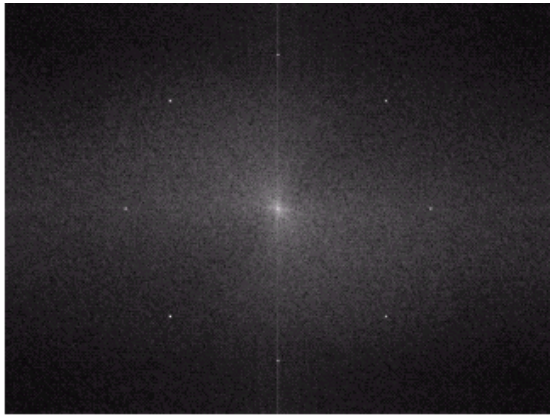
$$H(u,v)=1-e^{-\frac{1}{2}\left[\frac{D^2(u,v)-D_0^2}{D(u,v)W}\right]^2}$$



Bandreject Filters: Demo

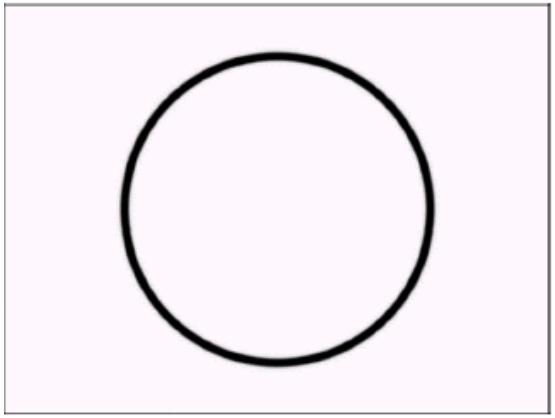
Original corrupted by sinusoidal noise





Fourier transform

Butterworth filter





Result of filtering

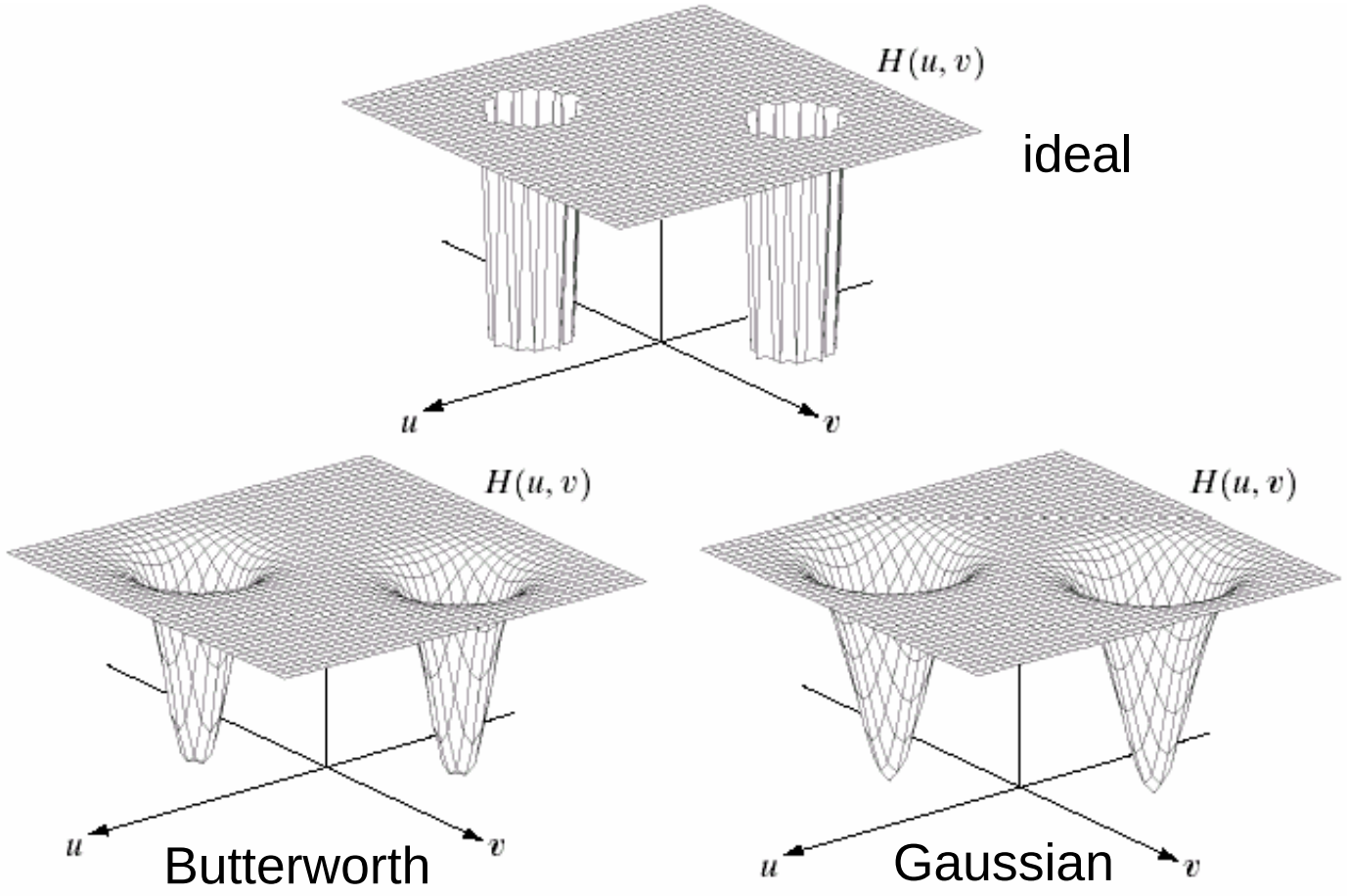
Notch Filters

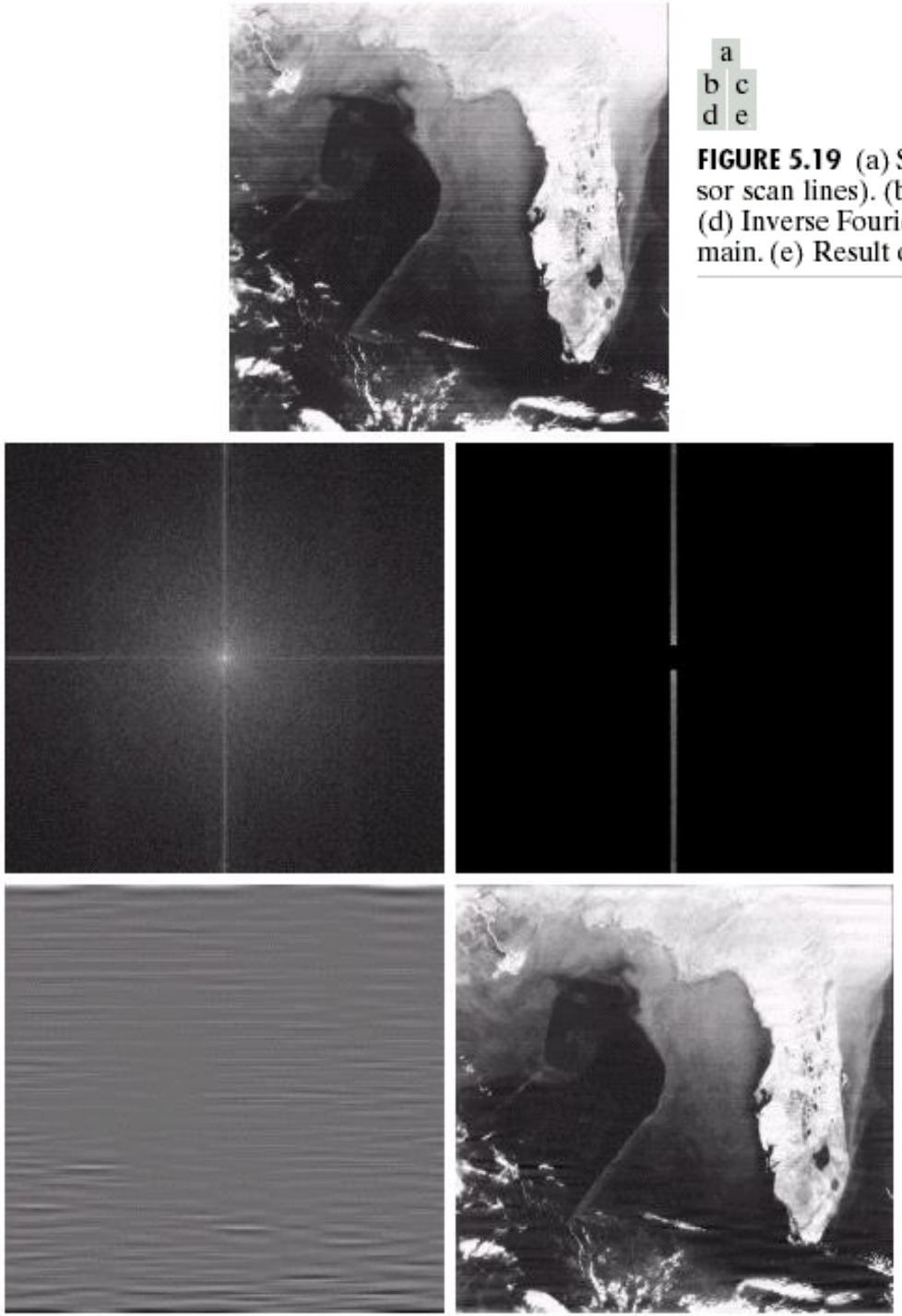
- Reject in predefined neighborhoods about the center frequency.
- Due to the symmetry of the Fourier transform, notch filters must appear in symmetric pairs about the origin.
- Given 2 centers (u_0, v_0) and ($-u_0$, $-v_0$), we define $D_1(u,v)$ and $D_2(u,v)$ as

$$D_1(u,v) = \left[(u - M/2 - u_0)^2 + (v - N/2 - v_0)^2\right]^{1/2}$$

$$D_2(u,v) = \left[(u - M/2 + u_0)^2 + (v - N/2 + v_0)^2\right]^{1/2}$$

Notch Filters: plots





a
b c
d e

FIGURE 5.19 (a) Satellite image of Florida and the Gulf of Mexico (note horizontal sensor scan lines). (b) Spectrum of (a). (c) Notch pass filter shown superimposed on (b). (d) Inverse Fourier transform of filtered image, showing noise pattern in the spatial domain. (e) Result of notch reject filtering. (Original image courtesy of NOAA.)

Reducing the effect of scan lines

Notch Filters (con'd)

- Ideal filter

$$H(u,v)=\begin{cases}0 & \text{if } D_1(u,v)\le D_0 \text{ or } D_2(u,v)\le D_0\\ 1 & \text{otherwise}\end{cases}$$

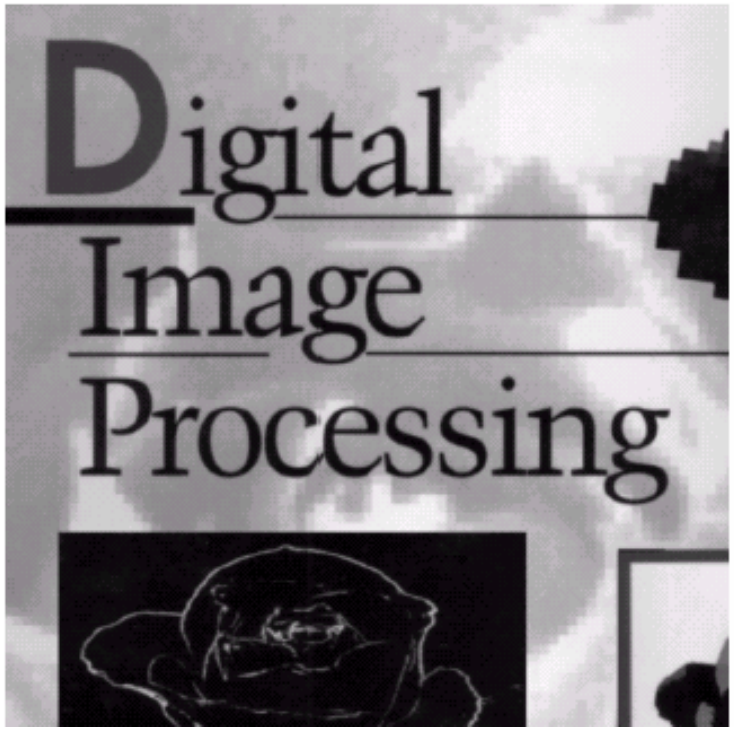
- Butterworth filter

$$H(u,v)=\frac{1}{1+\left[\dfrac{D_0^2}{D_1(u,v)D_2(u,v)}\right]^n}$$

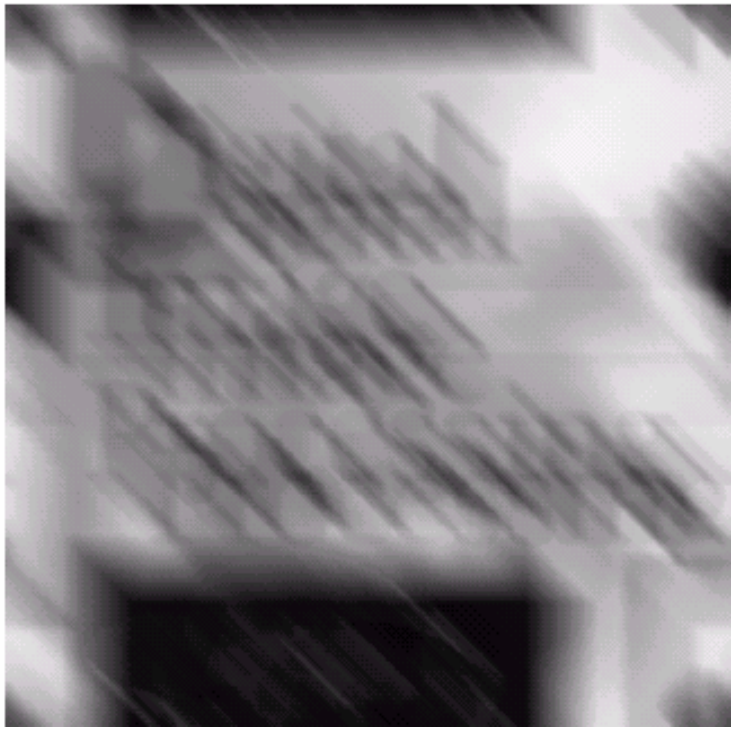
- Gaussian filter

$$H(u,v)=1-e^{-\frac{1}{2}\left[\frac{D_2(u,v)D_2(u,v)}{D_0^2}\right]}$$

How to deal with motion or out-of-focus blurring ?

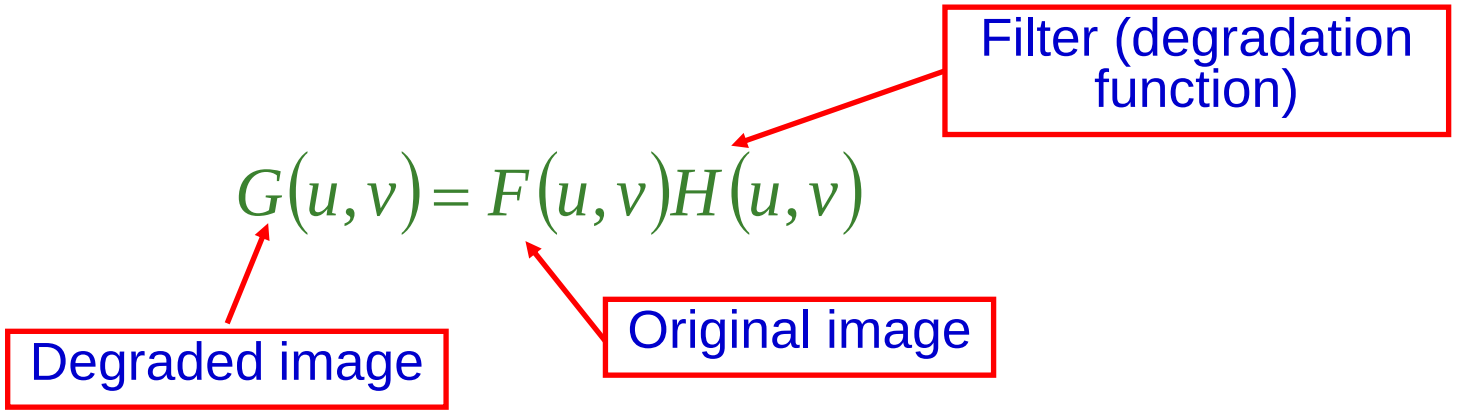


Original



Blurred by motion

Convolution Theory: Review



- Knowing the degradation function H(u,v), we can, in theory, obtain the original image F(u,v).
- In practice, H(u,v) is often unknown.
- We'll discuss briefly one method of obtaining the degradation functions. For interested readers, please consult Gonzalez, section 5.6 for other methods.

Estimation of $H(u,v)$ by Experimentation for out-of-focus

- If equipment similar to the one used to acquire the degraded image is available, it is possible, in principle, to obtain the accurate estimate of ***H(u,v)***.
  - Step1: reproduce the degraded image by varying the system settings.
  - Step2: obtain the impulse response of the degradation by imaging an impulse (small dot of light) using the same system settings.
  - Step3: recalling that FT of an impulse is a constant (A)

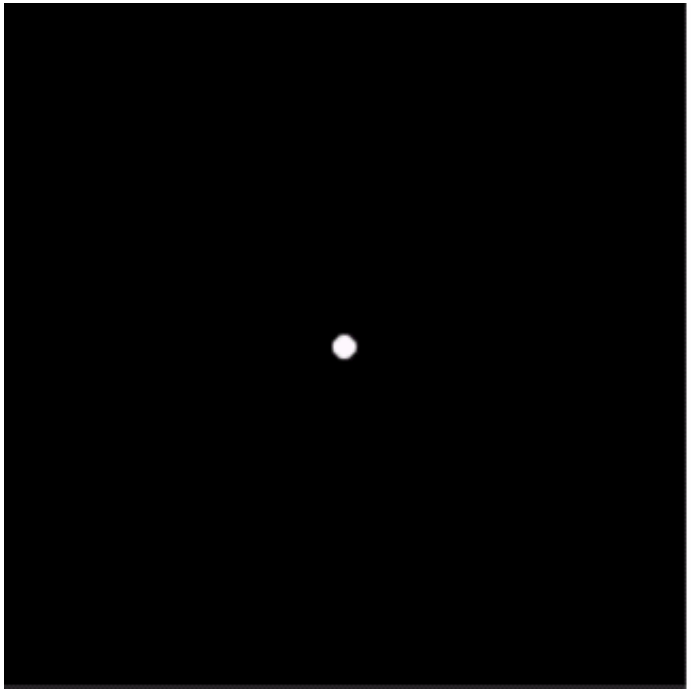
$$H(u,v) = \frac{G(u,v)}{A}$$

Degraded impulse image

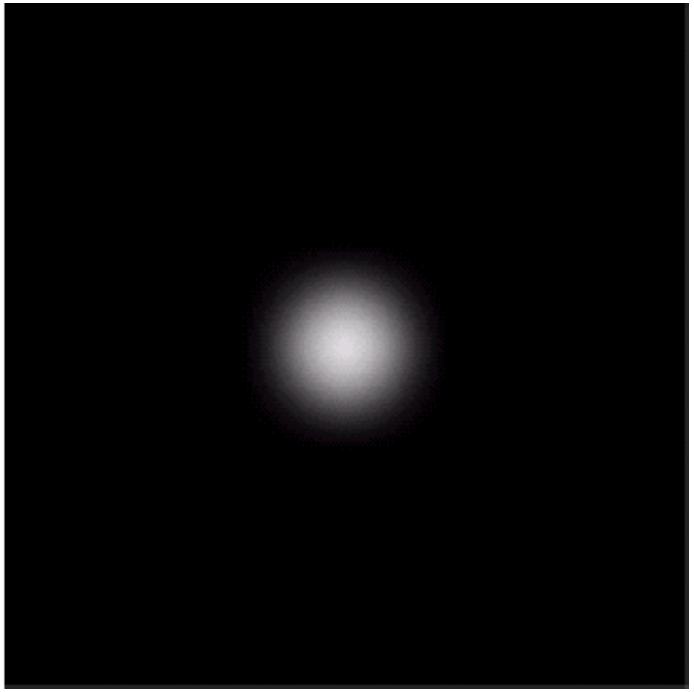
What we want

Strength of the impulse

Estimation of *H(u,v)* by Exp (con'd)

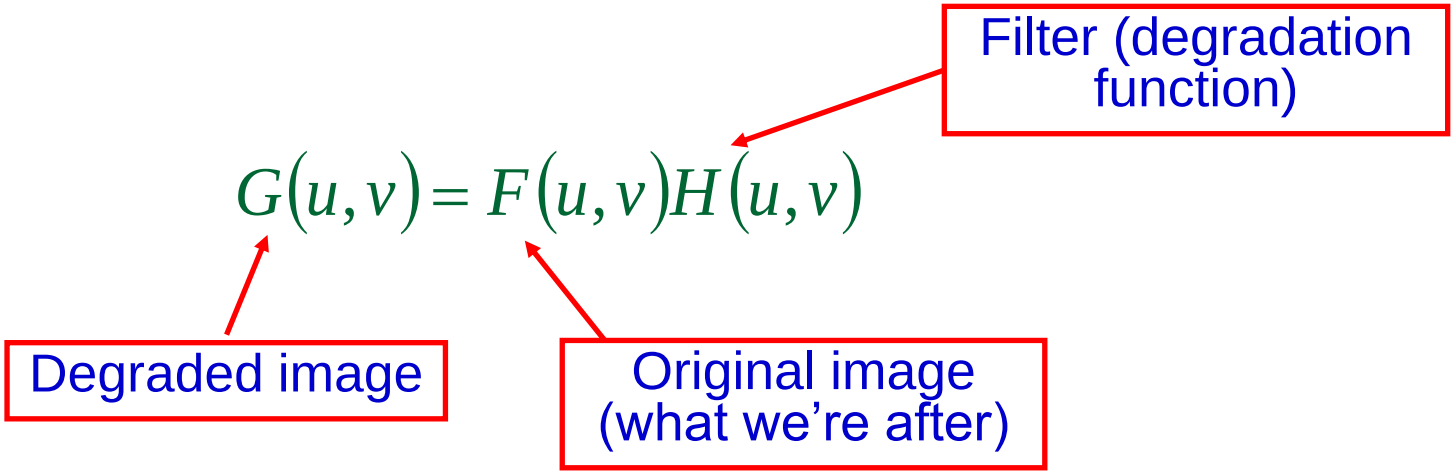


An impulse of light (magnified). The FT is a constant A



G(u,v), the imaged (degraded) impulse

Undoing the Degradation



- Knowing G & H, how to obtain F?
- Two methods:
  - Inverse filtering
  - Wiener filtering

Inverse Filtering

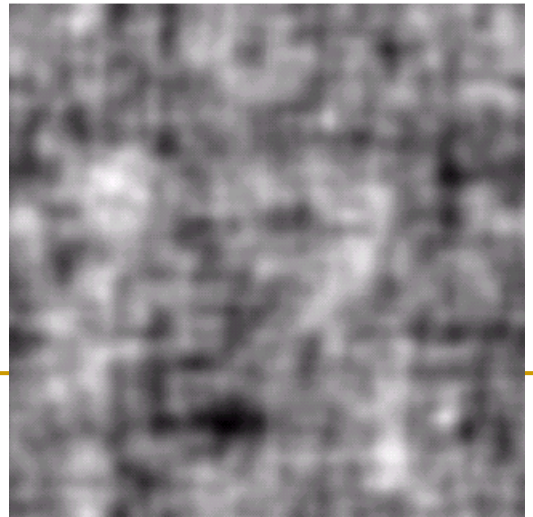
- In the simplest form:

Noise – random function

$$\hat{F}(u,v) = \frac{G(u,v)}{H(u,v)} = F(u,v) + \frac{N(u,v)}{H(u,v)}$$

- See any problems?
  - Division by small values can produce very large values that dominate the output.





Inverse Filtering (con'd)

- Solutions?
- There are two similar approaches:
  - Low-pass filtering with filter L(u,v):

$$F(u,v) = \frac{G(u,v)}{H(u,v)} L(u,v)$$

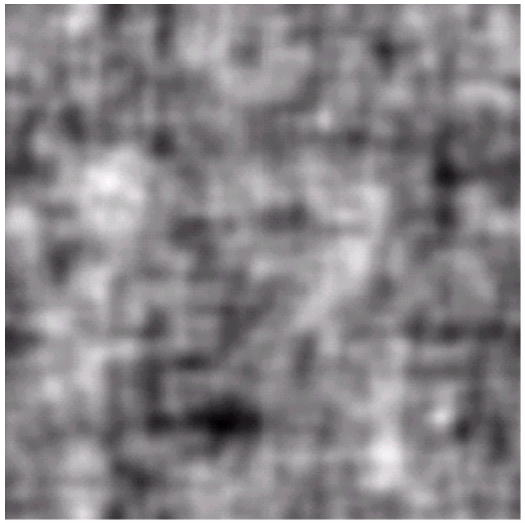
  - Thresholding (using only filter frequencies near the origin)

$$F(u,v) = \begin{cases} \dfrac{G(u,v)}{H(u,v)} & \text{if } |D(u,v)| \le d \\ G(u,v) & \text{if } |D(u,v)| > d \end{cases}$$

D(u,v) being the distance from the center

Inverse Filtering: Demo

Full filter

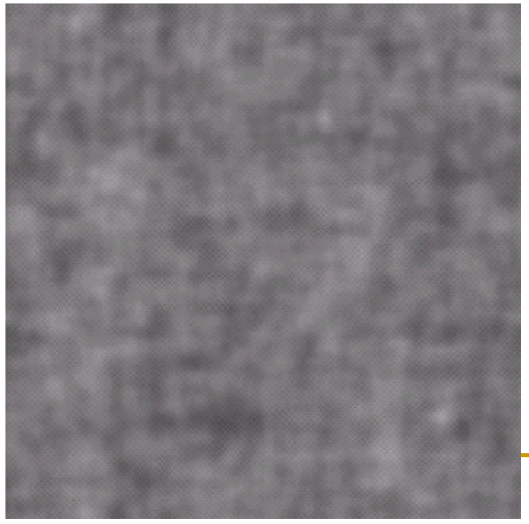




d=40

d=70





d=85

Inverse Filtering: Weaknesses

- Inverse filtering is not robust enough.
  - Doesn't explicitly handle the noise.
- It is easily corrupted by the random noise.

$$G(u,v) = F(u,v)H(u,v) + N(u,v)$$

$$=> F(u,v) = \frac{G(u,v) - N(u,v)}{H(u,v)}$$

- The noise can completely dominate the output.

Wiener Filtering

- What measure can we use to say whether our restoration has done a good job?
- Given the original image ***f*** and the restored version ***r***, we would like ***r*** to be as close to ***f*** as possible.
- One possible measure is the sum-squared-differences $\sum\left(f_{i,j}-r_{i,j}\right)^2$
- Wiener filtering: minimum mean square error: $\hat{F}(u,v)=\left[\frac{1}{H(u,v)}\frac{\left|H(u,v)^2\right|}{\left|H(u,v)\right|^2+K}\right]G(u,v)$

Specified constant

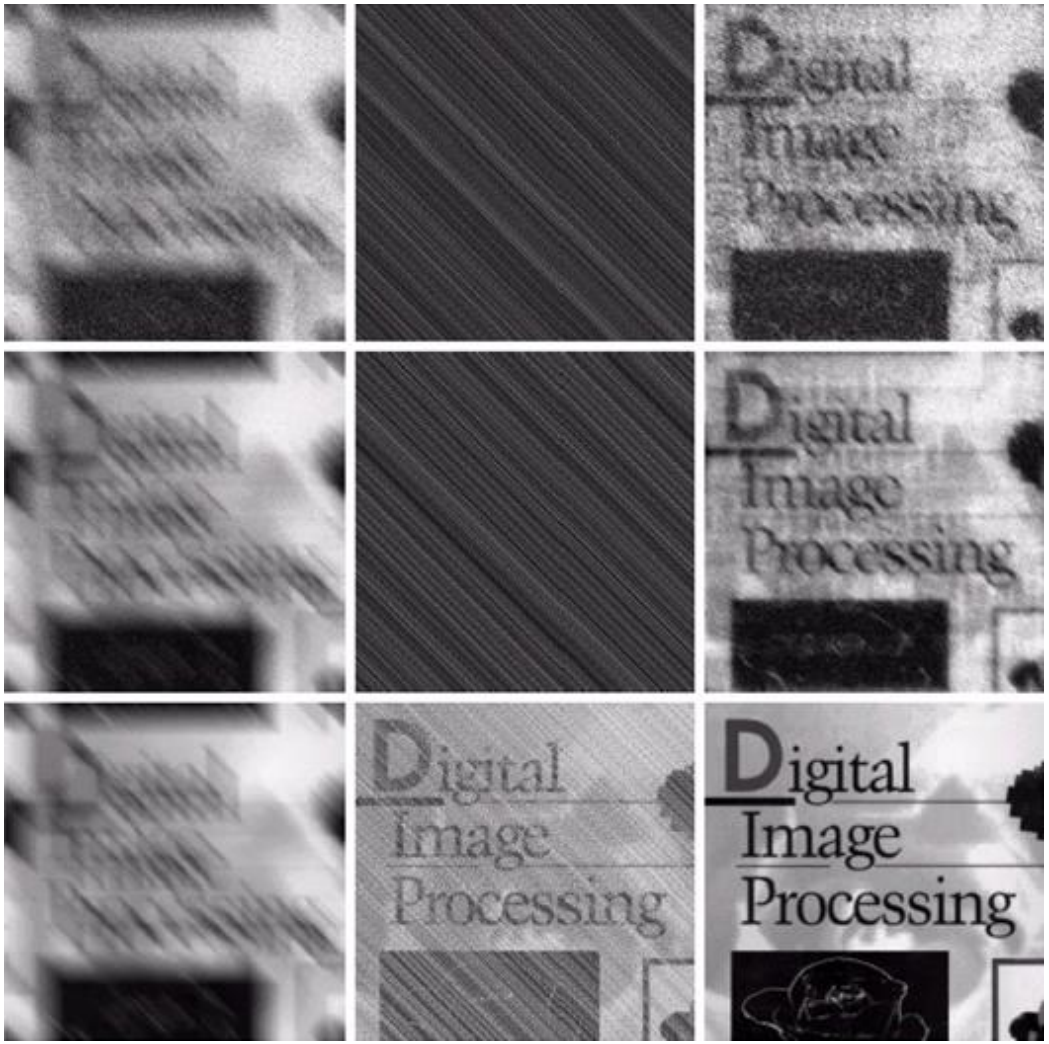
Comparison of Inverse and Wiener Filtering

- Column 1:

  blurred image with additive Gaussian noise of variances 650, 65 and 0.0065.
- Column 2:

  Inverse filtering
- Column 3:

  Wiener filtering



Summary

- Removal of periodic noise:
  - Bandreject
  - Notch filter
- Deblurring the image:
  - Inverse filtering
  - Wiener filtering